Galvanization Device Automation

Report Part 1: PLC Control Logic

Report Part 2: HMI Interface

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0.1 Introduction

This report describes the implementation of a galvanization device, including both its PLC-based automation logic and an accompanying HMI interface. The system is designed to transport a workpiece along a production line, perform material recognition, and apply a specified coating sequence according to the detected material. Additional safety measures, error handling, and manual control options are also integrated.

The implementation is divided into two main parts:

- PLC automation logic (implemented on Target 3).
- HMI interface (implemented on Target 5).

Only the high-level design is covered here; the source code and comments in the targets provide further detail.

The implementation needed to address the following aspects:

- Material recognition via inductive and optical sensors.
- Coating procedures for different materials.
- Safety measures and failure states.
- Manual control (both physical panel and HMI-based).

The following assumptions underlie the design:

- The stations are always visited in order.
- Every station has a sensor that indicates when the machine is in position.
- If the respective sensor triggers, the machine is considered in position to begin the coating.
- There are sensors at the start and end of the production line.
- There are sensors at the start and end of the material-recognition path (these may overlap with the line-end sensors).
- If a sensor fails to trigger, the machine is assumed to be able to safely reach the end of the line, provided that only one device is present on that line at any time to avoid collisions.

The implementation does not use absolute positions; instead, progression is based on station sensors and finite-state-machine logic. This reduces the precision of some safety checks — for example, if a device fails to stop at a station, the error is detected only when it reaches the end of the line — but improves flexibility, as stations can be moved or reconfigured without changing the core logic.

0.2 Control System Architecture

The full automation logic is implemented on Target 3 using structured text and function blocks. The system consists of two main state machines and several supporting functions.

- **Main(OB1)** – The main procedure. Implements a high-level state machine that:
  - Moves the device to the collection point.
  - Waits for a material input.
  - Executes a material recognition trip.
  - Optionally performs a control trip.
  - Executes the appropriate coating procedure.
  - Takes the material off the production line and returns to the starting point.
- **TRIP(FB1)** – A smaller state machine used to execute an arbitrary trip through the stations. The calling logic specifies how long the device should stop at each station. If the stop time is 0 s, the station is skipped. The function block can terminate either with success (Q = 1) or an error (E = 1).
- **TRAVEL(FC1)** – A function that triggers the device to move in a specified direction until it reaches a station. Stations are identified by their respective sensor and a key (SWT0–3). This allows certain stations to be disabled for simulation of breakdowns.
- **ZYL(FC2)** – Controls a mechanical cylinder (up/down). The function commands the cylinder and then verifies that the corresponding sensor (ZEndUp / ZEndDown) is active; otherwise, it raises an error.
- **WAIT(FC3)** – Uses a timer block to wait for a specified duration. Once the time expires, it resets the timer and transitions to the next state.

All function blocks assume that "progression" means moving to the immediately following state. This is a code-structure assumption rather than an environment constraint, so it can be easily maintained.

Further details are documented in the inline comments of the program.

0.2.1 Control Panel Elements

The following table describes the physical control panel elements and their functions:

Element	Manual Handling	Function
LED0	O	Vorbehandlung status (0 – taken or turned off)
LED1	O	Nickel electrolyte status (0 – taken or turned off)
LED2	O	Gold electrolyte status (0 – taken or turned off)
LED3	O	Rhodium electrolyte status (0 – taken or turned off)
LED4	O	0 if the cylinder position is invalid
LED5	O	Control-trip status (1 – control trip completed)
LED6	O	Material inductivity (1 – material is inductive)
LED7	O	Material reflectivity (1 – material is reflective)
SWT0–3	O	Enable/disable the respective station
SWT4	✓	Trigger cylinder (1 – cylinder down)
SWT5	×	Set control trip (see description)
SWT5	✓	Trigger ejection magnet (if key is on and PosELM is 0)
SWT6–7	×	Set IND and OPT flag values
SWT6–7	✓	Sensor reset (toggle from 0 to 1 to reset sensor)

Note: SWT0–3 should be set to 1 before starting a coating procedure.

0.2.2 Material Recognition

After signaling a start (e.g., via a green button), the program first performs a recognition trip. Instead of stopping at each sensor, the machine traverses the entire line, ensuring that both inductive (IndNS) and optical (OptNS) sensors are passed. The two properties measured are:

- Inductivity (IND)
- Reflectivity (OPT)

Recognition is based on a truth table:

Material	IND	OPT
Aluminum	1	1
Steel	1	0
Plastic (bright)	0	1
Plastic (dark)	0	0

Note 1: The inductivity sensor provides both a measurement and a validity signal. The implementation uses the negated validity signal for IND, as only inductive materials disturb the sensor. This approach also allows detection from a greater distance, especially for aluminum. However, this behavior may not generalize to all materials and can be adjusted if needed.

Note 2: After a recognition trip, the machine stops at the end of the line until user input is given. This allows verification of the sensor readings. The user has three options:

1. Readings are correct: Press the yellow button to proceed with coating.
2. A manual coating procedure is desired: Use SWT5–7 to set the control-trip, IND, and OPT flags manually; then press the white button, followed by the yellow button.
3. An error has occurred: Enter manual handling mode.

0.2.3 Coating Procedures

A coating procedure is selected automatically based on the recognized material and the following table:

Material	Vorbehandlung	Nickel	Gold	Rhodium
Plastic (dark)	2 s	1 s	1 s	–
Plastic (bright)	–	1 s	2 s	–
Steel	–	1 s	1 s	2 s
Aluminum	1 s	2 s	–	2 s

These correspond to the procedure variants described in the task specification. It is still possible to manually select a different procedure by setting the sensor flags (IND / OPT) after the recognition trip.

Each station is associated with a sensor:

- Vorbehandlung → MagE_D
- Nickel → KapNS
- Gold → PosELM
- Rhodium → MagE_U

The start of the line serves as both the collection point and the Vorbehandlung station.

0.2.4 Safety Measures

The following safety functions are implemented in the PLC logic:

- If the machine reaches the end of the track without stopping at a designated station, an error is raised:

$$\neg\texttt{goal} \wedge ((\texttt{MCDir} \wedge \texttt{MagE_D}) \vee (\neg\texttt{MCDir} \wedge \texttt{MagE_U})) \Rightarrow \texttt{Error} \tag{1}$$

$$\texttt{Error} \Rightarrow \neg\texttt{Verletzung} \tag{2}$$

  where `goal` is an internal flag indicating that an intended station was reached.

- When the cylinder is moved, the system waits for the corresponding sensor (ZEndDO / ZEndUP) to confirm position, preventing material loss during movement:

$$(\neg \texttt{ZEndDO} \wedge \texttt{ZylDO}) \vee (\neg \texttt{ZEndUP} \wedge \texttt{ZylUP}) \Rightarrow \neg \texttt{MCGo} \tag{3}$$

- If a travel command is issued while no piece is detected in the machine, an error is generated:

$$\neg \texttt{GlasLs} \Rightarrow \neg \texttt{MCGo} \wedge \texttt{Error} \tag{4}$$

- If the entry laser (EinLS) is interrupted, the process is paused until a yellow button or similar HMI interaction is used. This does not affect manual handling or the error state:

$$\neg \texttt{EinLS} \wedge \texttt{OldEinLS} \wedge \texttt{MCGo} \Rightarrow \texttt{Verletzung} \tag{5}$$

  In the current implementation:

$$\texttt{OldEinLS} \Rightarrow \neg \texttt{MCGo} \Rightarrow \neg \texttt{Verletzung} \tag{6}$$

0.2.5 Failure States

Both state machines include a dedicated failure state (state 100). When entered, all movement is stopped and the system becomes unresponsive. The intended recovery path is to switch to manual control and, if necessary, perform maintenance.

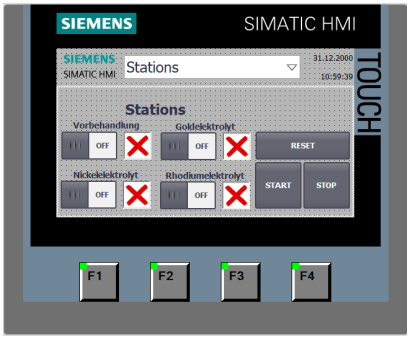
Manual Handling

Manual handling is implemented in state 15 of the main state machine. It can be triggered from any state — including the failure state — by either moving a joystick up/down or pressing a red button. In manual handling mode, the user can control:

- Movement along the track
- The cylinder position
- Sensor readings (OPT / IND, though these are effectively bypassed)
- The ejection magnet
- Exiting manual mode

Unlike automatic operation, where the device moves at full speed, manual mode allows the user to adjust speed via the joystick (MCSpeed). To exit manual handling, the red button is pressed again; the control-trip flag is reset and the program restarts from state 0.

Note: In this mode, the sensor readings IND and OPT are logically combined with SWT6–7 using an AND. To reset the readings, briefly set these switches to 0 and then back to 1.



0.3 HMI Interface Design

The HMI interface is implemented on Target 5 and provides visualization and control of the galvanization process. The system assumes familiarity with the PLC logic as described above, so internal PLC behavior is not repeated in detail where it is unchanged. The HMI consists of three main components:

- Visualization and control of process variables.
- HMI-triggered manual handling.
- Basic alarm handling.

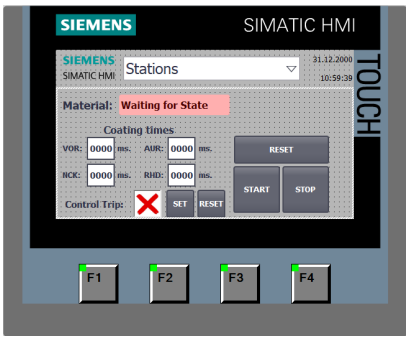
Due to time constraints and personal circumstances, two aspects were not completed to the intended standard: historical data logging / trends and thorough bug testing. The implementation is considered functional but would benefit from further refinement and additional testing.

The following assumptions guide the HMI design:

- Automatic processes can be controlled entirely from the HMI.
- Manual operations should prefer the physical control panel.
- The operator can visually observe the machine.

Under these assumptions, all switch/button functions have been moved to the HMI. The physical switches S0–4 are disabled, while the buttons remain functional and are largely interchangeable with their HMI equivalents in terms of machine behavior.

0.3.1 Automatic Process Screens



The top menu is retained as provided. Although it was considered to remove it and control the HMI entirely via buttons, the current setup offers slightly more control without introducing usability issues. The switches S0–4 are replaced by HMI-based toggles that are functionally equivalent to the switches in the first protocol. They allow certain stations to be deactivated to simulate breakdowns or maintenance. Graphical I/O indicators next to each station show its current status:

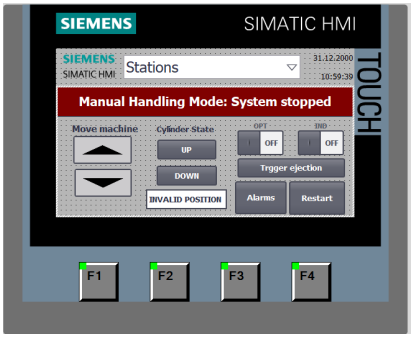
✓	Station active and available
O	Station active but occupied
X	Station inactive / broken (switch off)

Each screen provides a RESET and STOP button. RESET immediately stops the current procedure and returns the system to the initial state. STOP immediately halts the process and switches the machine into HMI-based manual mode.

On the material recognition screen, the currently recognized material is displayed and can be changed manually. Once the machine is in the appropriate state, a dropdown menu appears at the top, with the currently detected material pre-selected. The operator can manually choose any of the four materials to start the corresponding coating procedure. The screen also shows the process times for each material as information. A future enhancement would be to add support for custom procedures, including user-defined process times. Below this, the current "control trip" status can be viewed and set manually. Pressing the START button initiates the corresponding trip.

0.3.2 Manual Control via HMI

For manual control, the system supports input from both the physical panel and the HMI. The physical panel control takes precedence over HMI-based manual handling:



- Physical panel manual handling > HMI manual handling

HMI-controlled manual handling is implemented in state 20 and is accessed via the following screen:

From this screen, the user can:

- Move the device along the track.
- Control the cylinder.
- Set sensor readings for OPT and IND (effectively for testing).
- Control the ejection magnet.
- Exit manual mode and return to the initial state.

Switching between screens is blocked while manual mode is active, to avoid inconsistent state transitions.

0.3.3 Alarm System

The system implements seven distinct alarms to cover the main critical and informative states:

- Emergency stop triggered (Error)
- Track laser interrupted (e.g., an object or person crossing the line) (Error)
- Station sensor failed to trigger when expected (Error)
- No piece detected in the machine (Warning)
- Cylinder failed to reach the expected position (Warning)

- Material was recognized or changed (Note)
- Trip was successfully completed (Note)

These alarms capture all important error states defined in the original PLC implementation, as well as the new emergency stop (Notaus) added by the HMI. The classification (Error / Warning / Note) helps the operator distinguish between safety-critical events, operational warnings, and purely informational messages.